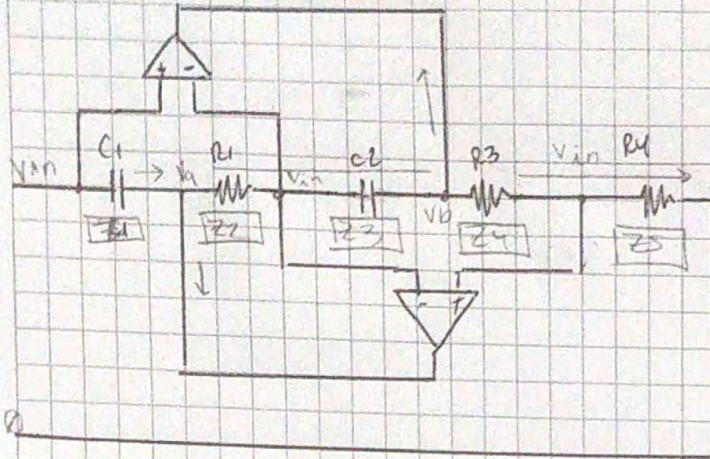


Ejercicio 8

→ De forma general, porque el $\neq$ es "igual"



$$(I) \frac{V_{in} - V_a}{Z_1} = I_{in}$$

$$V_{in} - V_a = I_{in} Z_1$$

$$V_{in} - I_{in} Z_1 = V_a$$

$$(II) \frac{V_a - V_{in}}{Z_2} = \frac{V_{in} - V_b}{Z_3}$$

$$(III) \frac{V_b - V_{in}}{Z_4} = \frac{V_{in} - 0}{Z_5}$$

$$V_b - V_{in} = V_{in} \frac{Z_4}{Z_5}$$

$$V_b = V_{in} \left(\frac{Z_4}{Z_5} + 1 \right)$$

→ junto todo en función de V_{in}

$$(V_a - V_{in}) Z_3 = (V_{in} - V_b) Z_2$$

$$[(V_{in} - I_{in} Z_1) - V_{in}] Z_3 = \left\{ V_{in} - \left[V_{in} \left(\frac{Z_4}{Z_5} + 1 \right) \right] \right\} Z_2$$

$$-I_{in} Z_1 Z_3 = V_{in} Z_2 - V_{in} \frac{Z_2 Z_4}{Z_5} - V_{in} Z_2$$

$$I_{in} Z_1 Z_3 = V_{in} \frac{Z_2 Z_4}{Z_5}$$

$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{Z_1 Z_3 Z_5}{Z_2 Z_4}$$

$$Z_{in} = \frac{1}{\frac{1}{R_1} \cdot \frac{1}{R_2} \cdot R_4} = \frac{R_4}{R_1 R_2}$$

$$Z_{in} = \frac{R_4}{\frac{1}{\frac{1}{R_1} \cdot \frac{1}{R_2} \cdot R_4}} = \frac{1000}{\frac{1}{\frac{1}{1000} \cdot \frac{1}{1000} \cdot 1000}} = \frac{1000}{\frac{1}{1000}} = 1000$$

